
TCX2005: Information Systems, Management and Organisations

L4: IS, Organizations, & Strategy-II

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Lesson Objectives



The Business Value Chain Model



Build v/s Buy: System acquisition strategy



PESTLE Analysis



What are the competition strategies for organizations to handle disruptive technology and disruptors?

Learning Resources



- Textbook (read before class)
 - Read chapter 3
- Any Other Material
 - Shared on Canvas for this week

Case Study

Read the case study from textbook on –Singapore as a Smart Nation

Identify:

- a. The problem(s)
- b. The solution
- c. Outcomes
- d. Key learning point(s)

Note: be ready to share your thoughts

Case Study

Problem:

- Pressures of increasing urban density
- Aging population
- Limited room to expand transport infrastructure
- Needs to consider the climate in housing planning

Solution: SMART Nation Initiative

Plan to use IT/IS for transformation through:

- Data collection sensors
- Large scale data centers
- National artificial intelligence strategy
- Smart infrastructure projects:
 - Realtime Singapore Traffic Watch
 - Smart Health Video Consultation
 - Automated Meter Reading
 - Pneumatic Waste Conveyance System

Transportation

- Traffic alerts and indexes
- Surveillance cameras
- Electronic Road Pricing System
- Testing driverless vehicles

Healthcare

- Video consultation system
- TeleRehab kit
- RoboCoach for seniors
- Drone survey mosquito breeding hotspots

Case Study

Outcome:

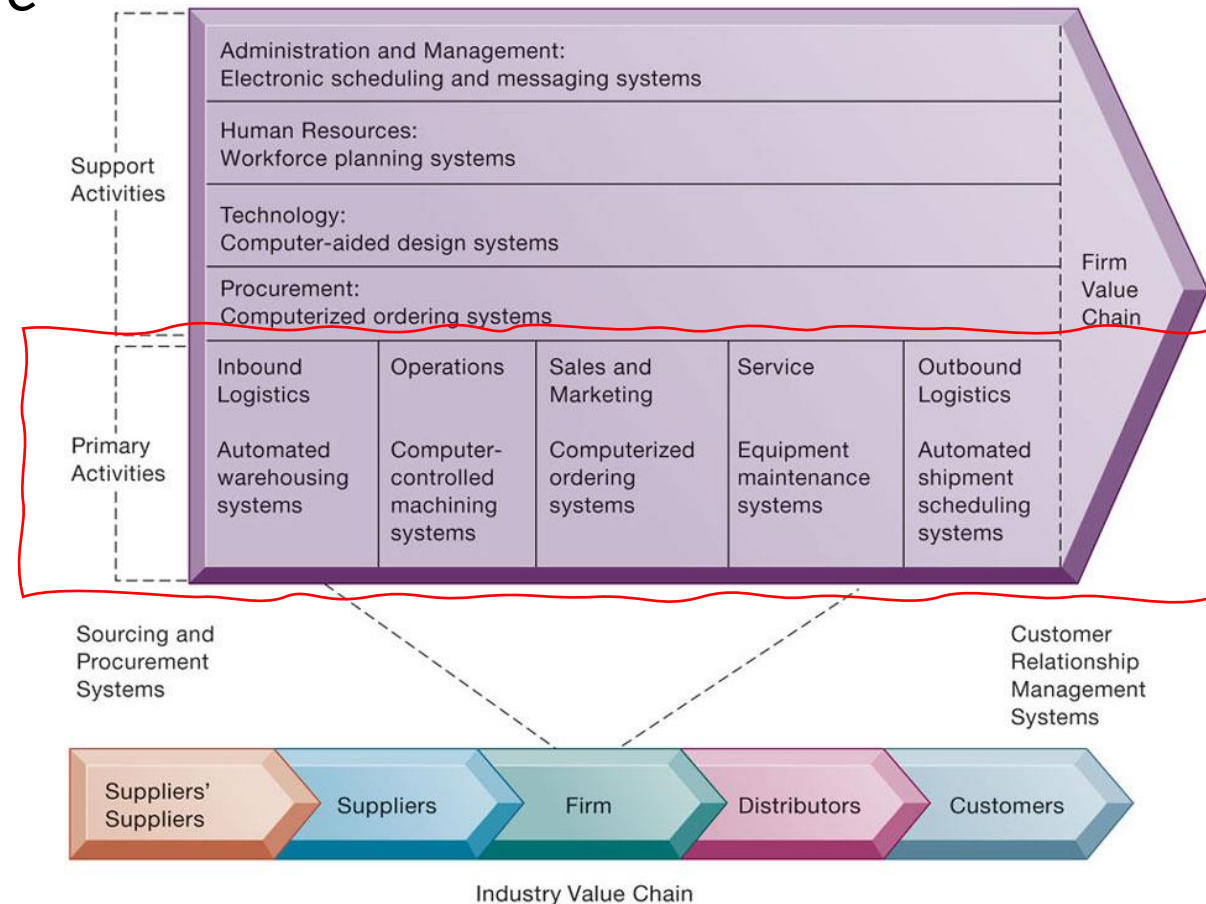
- Named Smart City of 2018
- High-speed internet coverage and high smartphone penetration
- Successfully involving people, private sector, and government agencies

Learning:

- Smart nation initiatives require significant implementation time
- Technology adoption needs to consider all citizen groups (elderly, less tech-savvy etc.)
- IS (High-speed internet and smartphone) enable service delivery
- Integration of multiple IS (traffic, healthcare, housing) is crucial
 - When systems are connected, data is available for AI and analytics to support decision-making (urban planning, traffic)
- Privacy and security issues must be addressed (remember TCX1003?!)

The Business Value Chain Model

- Firm as series of activities that add value to products or services
- Highlights business activities where competitive strategies can best be applied
 - **Primary** activities vs. **Support** activities
- At each stage, determine how information systems can improve operational efficiency and improve customer and supplier intimacy
- Utilize benchmarking, industry best practices



Connection to Business Value Chain Model

- The Singapore Smart Nation initiative demonstrates the Value Chain Model principles:

Improving Operational Efficiency

- Automated Meter Reading for water usage
- Pneumatic Waste Conveyance System
- Electronic Road Pricing System

Enhancing Customer Service

- Smart Health Video Consultation
- TeleRehab kit for rehabilitation
- RoboCoach for senior fitness

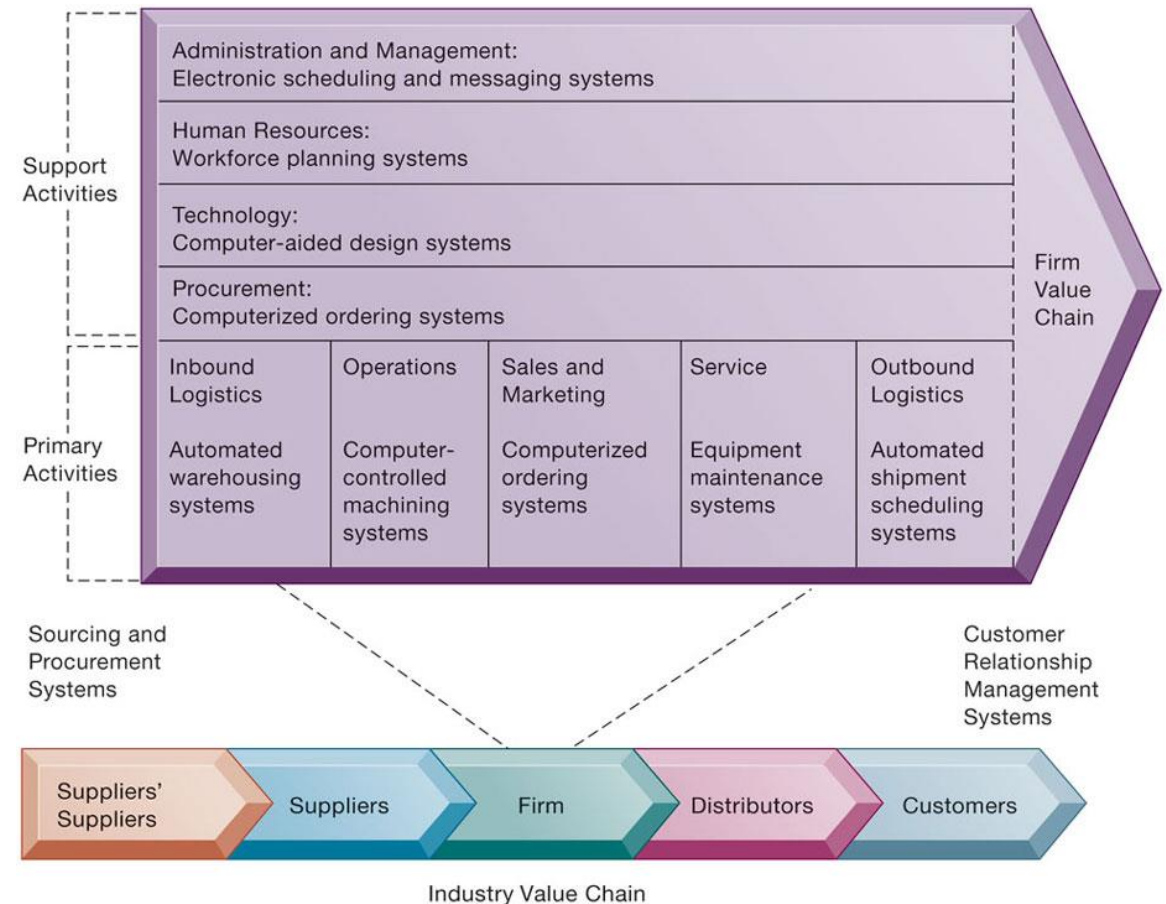
Information Systems for Planning

- Virtual Singapore for urban planning
- Traffic Watch for transport management
- CODEX platform for internal data sharing

From Value Chain Analysis to System Decisions

Value Chain Analysis leads to System Decisions

- Key steps
 1. **Analyze** value chain stages
 2. **Identify** where information systems can improve:
 - Operational efficiency
 - Customer intimacy
 - Supplier relationships
 3. **Generate** candidate applications
 4. **Prioritize** which to develop first



From Value Chain Analysis to System Decisions

System Acquisition Considerations

Leading to Build vs Buy

- How to improve business processes?
- What applications are needed?
- What solutions already exist?
- Which to build/buy first?

The Build v/s Buy Dilemma

Should we build or buy our information system?

Key Considerations

1. Strategic Value

- How unique are your business processes?
- Will a custom solution provide competitive advantage?

2. Resource Availability

- Do you have the necessary technical expertise (build & maintain)?
- What is your budget and timeline?

3. Market Solutions

- Are there existing solutions that meet your needs?
- How much customization would be required?
- What is the vendor ecosystem like?

4. Risk Assessment

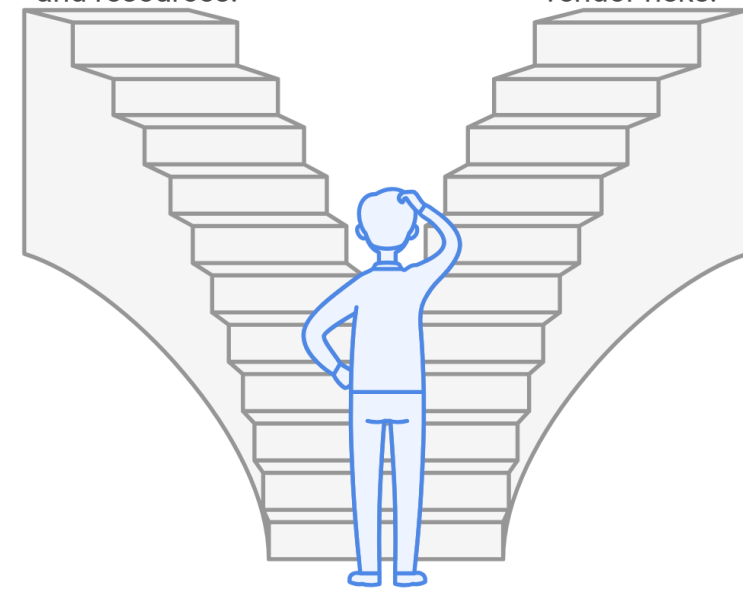
- Development risks (technical, timeline, budget)
- Vendor risks (stability, support, lock-in)
- Integration risks
- Security considerations

Build

Offers custom solutions and potential competitive advantage but requires expertise and resources.

Buy

Provides ready-made solutions with vendor support but may involve customization and vendor risks.



The Build v/s Buy Dilemma

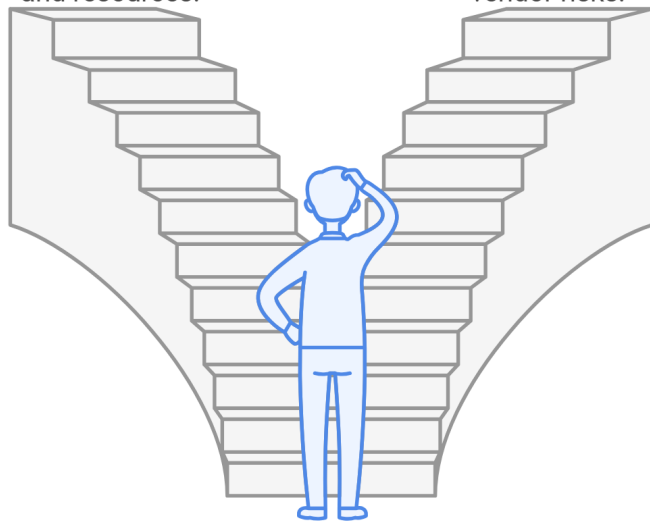
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How Singapore Airlines is building customer loyalty through blockchain

Feature

Jul 18, 2019

Blockchain

Technology Industry

Through the innovative use of blockchain services, the national carrier has created a groundbreaking loyalty programme

[source](#)

The Build v/s Buy Dilemma

Common Scenarios

- **Build** when unique business processes provide competitive advantage
- **Buy** when standardized processes are acceptable
- **Hybrid** approach: buy core system and build custom modules

The Build v/s Buy Dilemma

- **Hybrid** approach: buy core system and build custom modules

NEWS | This article is part of our Essential Guide: [Essential Guide: How APAC firms can ride out the pandemic](#)

How DBS is reaping the dividends of digital transformation

At DBS Bank, the Covid-19 coronavirus outbreak has demonstrated the value of technology and the investments it made over the past decade to modernise its technology stack



By Aaron Tan, TechTarget

Published: 22 Apr 2020 7:46

When some DBS Bank employees tested positive for Covid-19 early this year, the bank was able to use data from office access cards and Microsoft Office 365 calendars to conduct contact tracing within days. And within weeks, it rolled out an application that enables business customers to submit forms without visiting a branch.



[source](#)

System Acquisition Strategies

Internal acquisition

- **In-house development**
 - Complete control over development process and final product
 - Better alignment with specific business needs
 - Builds internal expertise and knowledge
- **Challenges**
 - Requires significant technical expertise
 - Time-consuming and resource-intensive
 - Higher development and maintenance costs
 - Risk of key personnel departure

External acquisition

- **Prepackaged system**
 - Commercial off-the-shelf (COTS) software package. E.g., MS Office 365
 - Cloud solutions (SaaS). E.g., Salesforce CRM
 - Lower initial costs and proven reliability
 - Regular updates and support
- **Outsourcing System Development**
 - Cost-effective access to specialized expertise
 - Focus on core business functions. E.g., Hiring a development firm to create a mobile app.
 - Questions to answer: what to outsource and when?

Decision Framework for Acquisition

Task Classification

Strategic Tasks

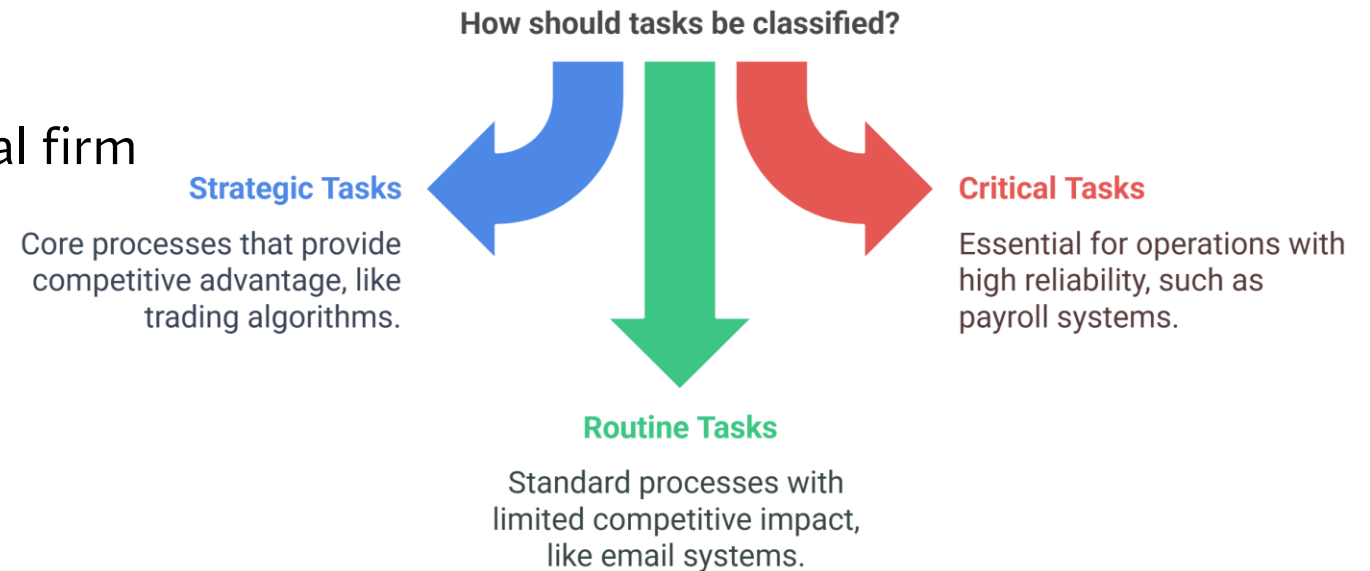
- Core business processes
- Competitive advantage
- E.g., Trading algorithms for a financial firm

Critical Tasks

- Essential for operations
- High reliability required
- E.g., Payroll system

Routine Tasks

- Standard business processes
- Limited competitive impact
- E.g., Email system



Decision Framework for Acquisition

Key Decision Factors for System Acquisition

1. Strategic Importance

- How critical is the system to your competitive advantage?

2. Cost Consideration

- Initial development/purchase costs plus any ongoing operational costs (licensing, maintenance, updates)
- Hidden costs (training, infrastructure upgrades, integration)

3. Quality Requirements

- Performance needs (speed, reliability (e.g., uptime), scalability)
- Security requirements

4. Regulatory Compliance

- Industry-specific regulations (HIPAA for healthcare, SOX for finance)
- Data protection requirements (GDPR, PDPA)

5. Time Constraints

- Project deadlines and market pressure

6. Risk Assessment

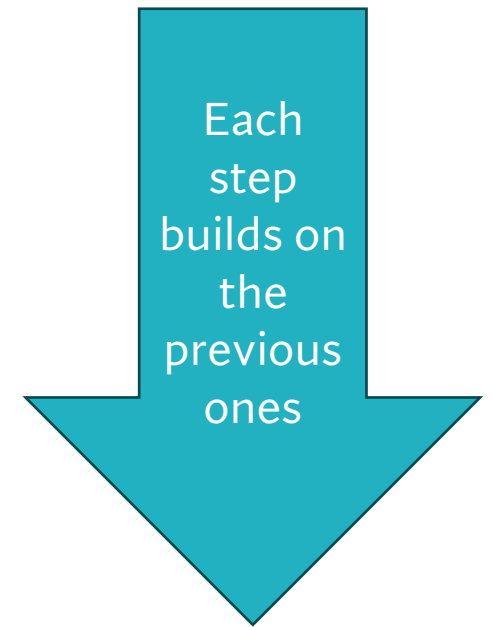
- Vendor stability and technology obsolescence

Etc.

External Acquisition Process

Key steps

1. Systems Planning and Selection
2. Feasibility Analysis
 - 5 dimensions: *organizational, political, technical, economic, and operational*
3. System Analysis
4. RFP Development
 - A document that is used to tell vendors what your requirements are and to invite them to provide information about how they might be able to meet those requirements.
5. Proposal Evaluation
6. Vendor Selection



External Acquisition Process

Key steps

Vendor Selection

- E.g.,

Hardware Criteria	Software Criteria	Other Criteria
Brand/manufacturer	Business alignment	Installation/training
Speed/storage	Required/desired features	Vendor characteristics (years in business, flexibility, reputation, etc.)
Reliability	Reliability	Price
Scalability for growth	Operations integration	
Ease of installation	Scalability for growth	
Ease of integration	Support model	
Warranty	Usability	

NUS Digital Transformation Journey

Learning Management System (LMS) Evolution

System Timeline

1. IVLE (1999-2018)

- In-house developed
- Served for 19 years



2. LumiNUS (2018-2022)

- On live since 2018.08
- In-house developed



3. Canvas

- On live since 2022.05
- External acquisition (SaaS)
- Current platform

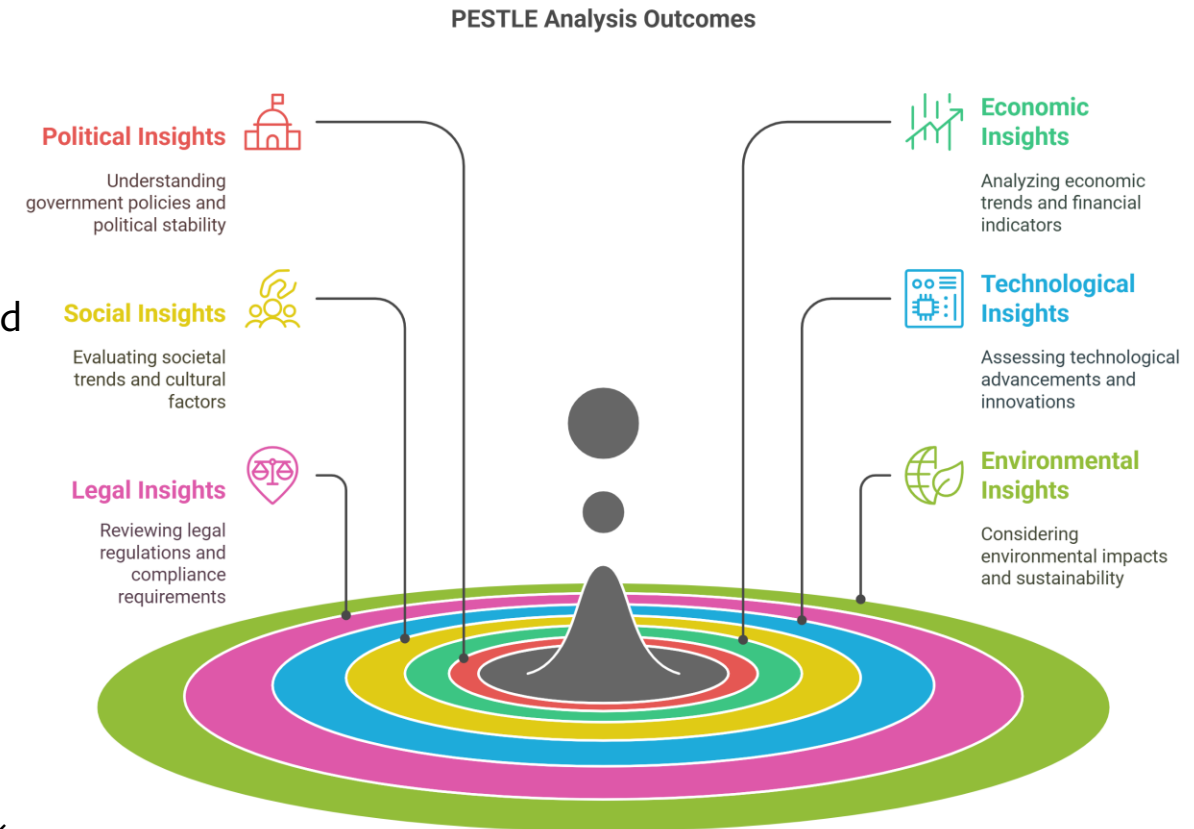


PESTLE Analysis

PESTLE Analysis

A framework for analyzing external factors that may affect IS acquisition decision and implementation success.

- Risk Management
 - Identifies external challenges early
 - Helps plan mitigation strategies
 - Ensures compliance
 - Helps assess political stability and policy changes that could impact implementation
- Better Decision Making
 - Provides structured evaluation framework
 - Considers factors beyond technical specifications
- Implementation Success
 - Anticipates potential adoption issues
 - Helps prepare change management plans
 - Helps align with national digital transformation initiatives (e.g., Smart Nation in Singapore)



Texas fails to change dealership law— Tesla still can't sell directly to state residents

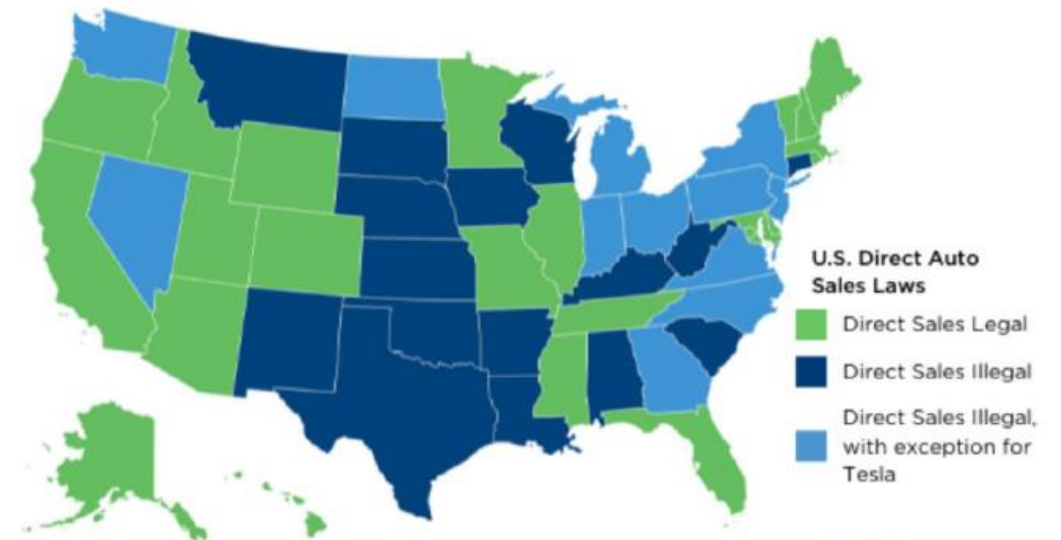
Posted May 28, 2021 by [Charles Morris](#) & filed under [Newswire](#), [The Vehicles](#).

For almost a decade now, Tesla has been fighting state governments for the right to sell its vehicles directly to customers—something which has historically been prohibited in most US states. Texas is one of the states that still forbids direct sales, so residents have had to resort to a work-around some call the “Texas two-step.” Texans can order a Tesla through the company’s web site, but no orders may be placed or processed within the state. The buyer must pay for the vehicle online, and can then pick it up at one of Tesla’s eight Texas service centers.

Source:

- <https://chargedevs.com/newswire/texas-fails-to-change-dealership-law-tesla-still-cant-sell-directly-to-state-residents/>
- <https://www.eesi.org/articles/view/beyond-cities-breaking-through-barriers-to-rural-electric-vehicle-adoption>

STATUS OF DIRECT AUTO SALES*



Graphic By: Jaxon Tolbert and Sydney O'Shaughnessy

*Direct auto sales laws by state as of 2021.

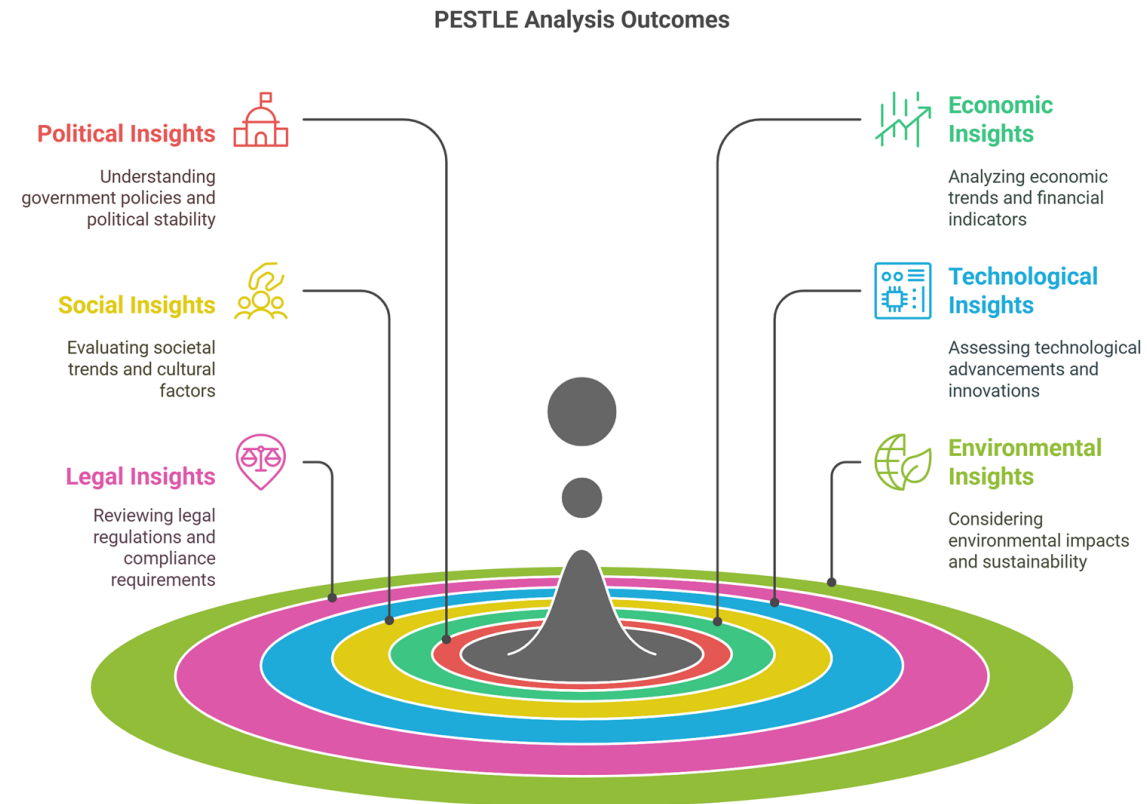


Class Activity: PESTLE analysis

Case Scenario: A large public hospital in Singapore is planning to acquire a new Electronic Health Records (EHR) system to improve patient data management, reduce paperwork, and ensure compliance with government regulations. The hospital must evaluate multiple external factors before proceeding with the acquisition to ensure a successful implementation.

Task: Using the provided PESTLE framework, analyze the external factors that may influence this IS system acquisition.

Hint: Consider government regulations, economic feasibility, user adoption, security risks, and other relevant factors.



PESTLE Analysis

- **Political**

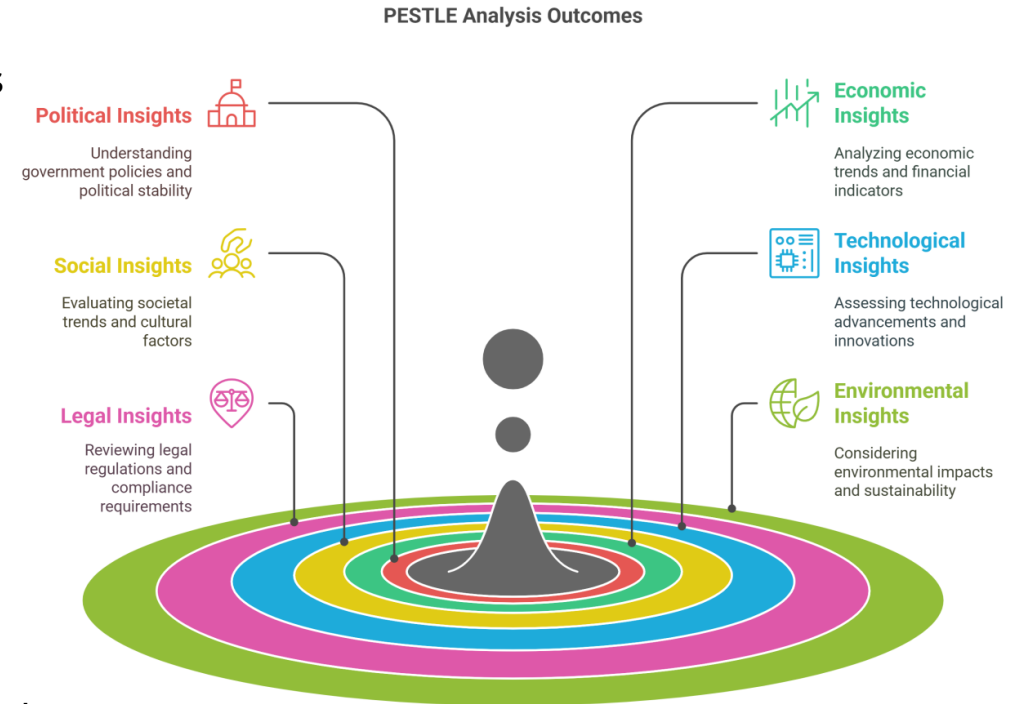
- Government IT policies such as data localization requirements
- *Example:* A hospital in Singapore ensures its new EHR system follows PDPA regulations to protect patient data.

- **Economic**

- Budget Constraints
- Return on Investment (ROI)
- *Example:* The hospital assesses if automation in the EHR system can reduce administrative costs in the long run.

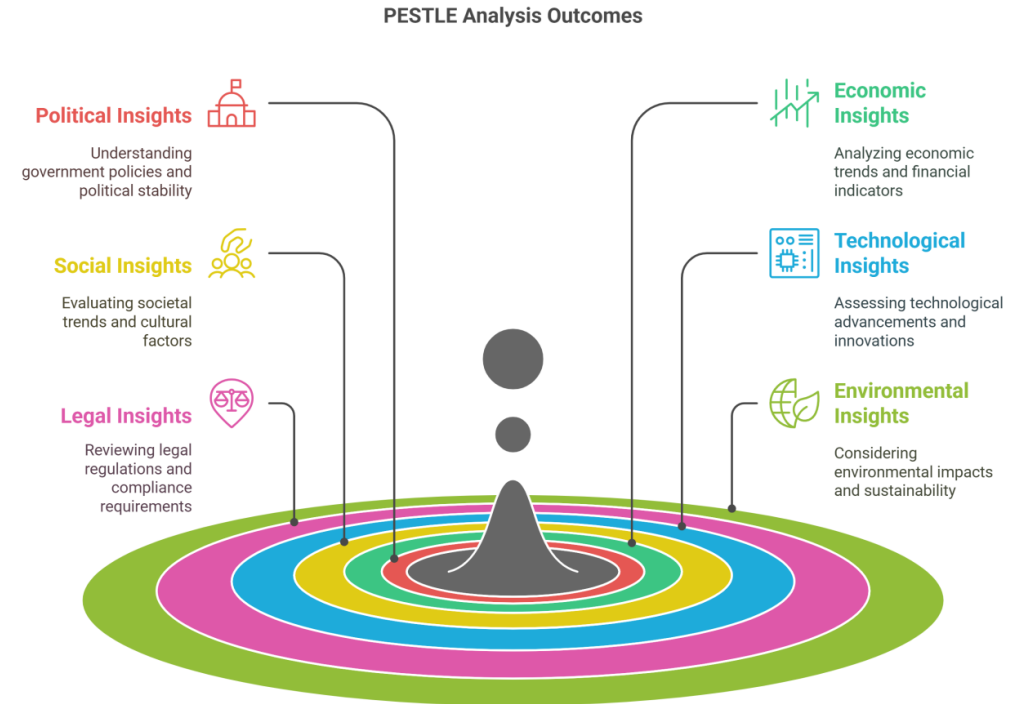
- **Social**

- User demographics and preferences (user acceptance)
- Digital literacy levels and/or training needs
- *Example:* The hospital conducts workshops for staff before launching the system to ensure smooth adoption.



PESTLE Analysis

- Technological
 - System Compatibility
 - Cybersecurity Measures
 - *Example:* The hospital ensures the EHR system integrates with existing medical imaging software (MRI, X-rays, etc.).
- Legal
 - PDPA requirements
 - Industry regulations
 - *Example:* Healthcare systems must comply with specific MOH data protection guidelines.
- Environmental
 - Green computing initiatives
 - Sustainability goals
 - *Example:* The hospital reduces its paper usage by 70% by switching to electronic patient records.



Disruptive Technologies

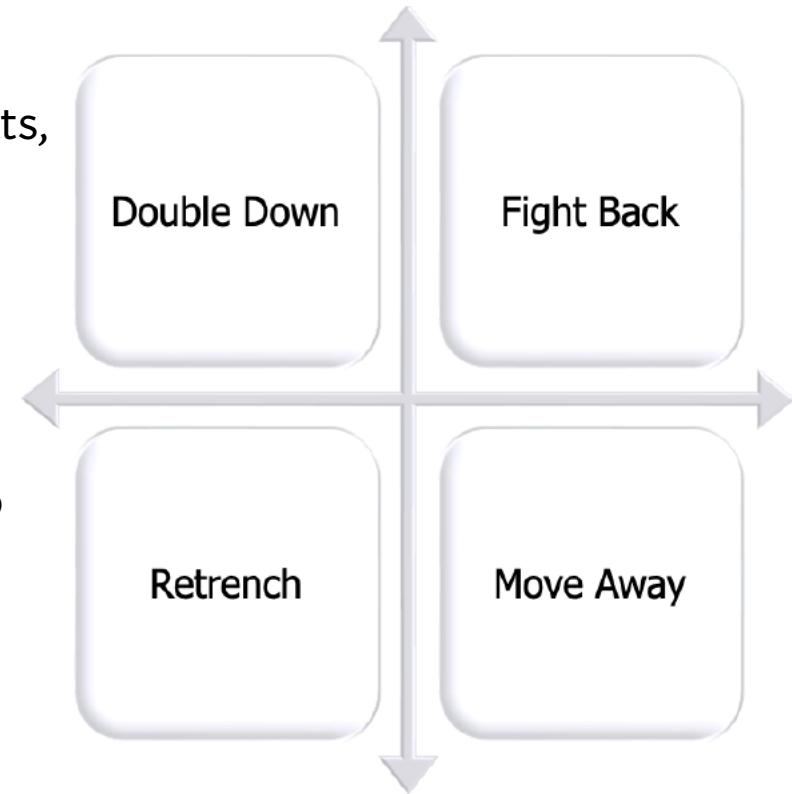
- Examples
 - Internet
 - Digital photography
 - Smartphone
 - IoT
 - Mobile payments
 - 3D printing
 - AI/ML
 - Blockchain
 - Generative AI

Disruptive Technologies

- What makes it “disruptive”?
 - Substitute products that perform as well as or better (often much better) than existing product
 - Technology that brings sweeping change to businesses, industries, markets
 - First movers and fast followers
 - First movers—inventors of disruptive technologies
 - Fast followers—firms with the size and resources to capitalize on that technology

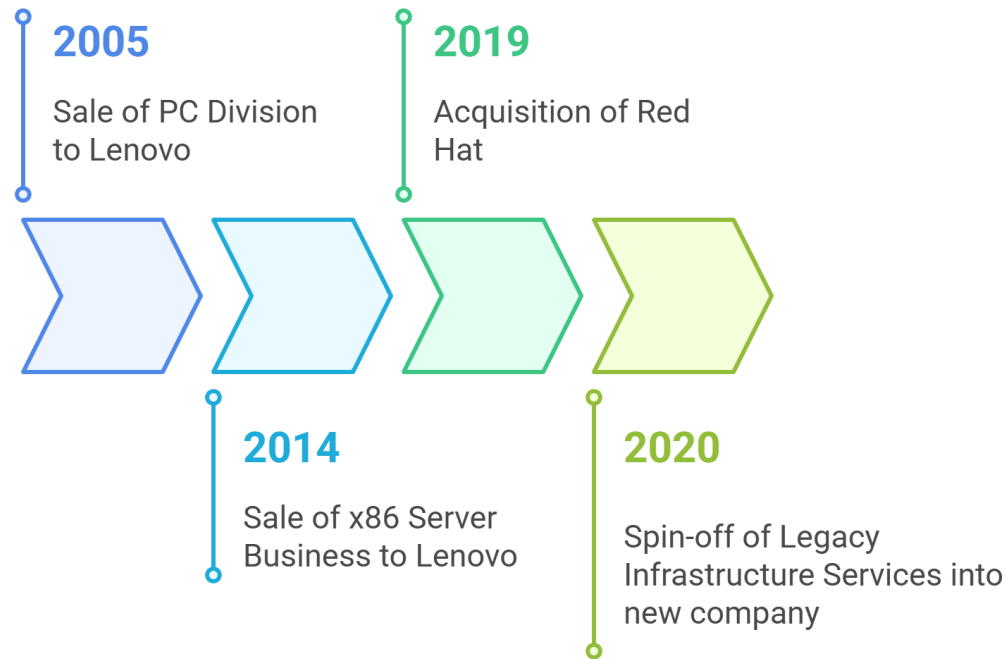
Adapting to Digital Disruption

- Adaptation strategies
 - Fight back
 - Incumbents go head-to-head against disrupters by setting up new units, making an acquisition, or entering a joint venture.
 - E.g., major carmakers' moves into electric vehicles
 - Double down
 - Incumbents focus on existing strengths.
 - Retrench
 - Incumbents yield ground to new arrivals and use a variety of tactics to ensure their own continued survival.
 - E.g., consolidation through mergers and acquisitions; seek help from government and regulators in putting additional constraints on new entrants
 - Move Away
 - Incumbents migrate to new opportunities



Adapting to Digital Disruption

- Example: IBM's Strategic Shift



Made with  Napkin

IBM: Mental Models in Disruption

Case Study



Naresh Sekar

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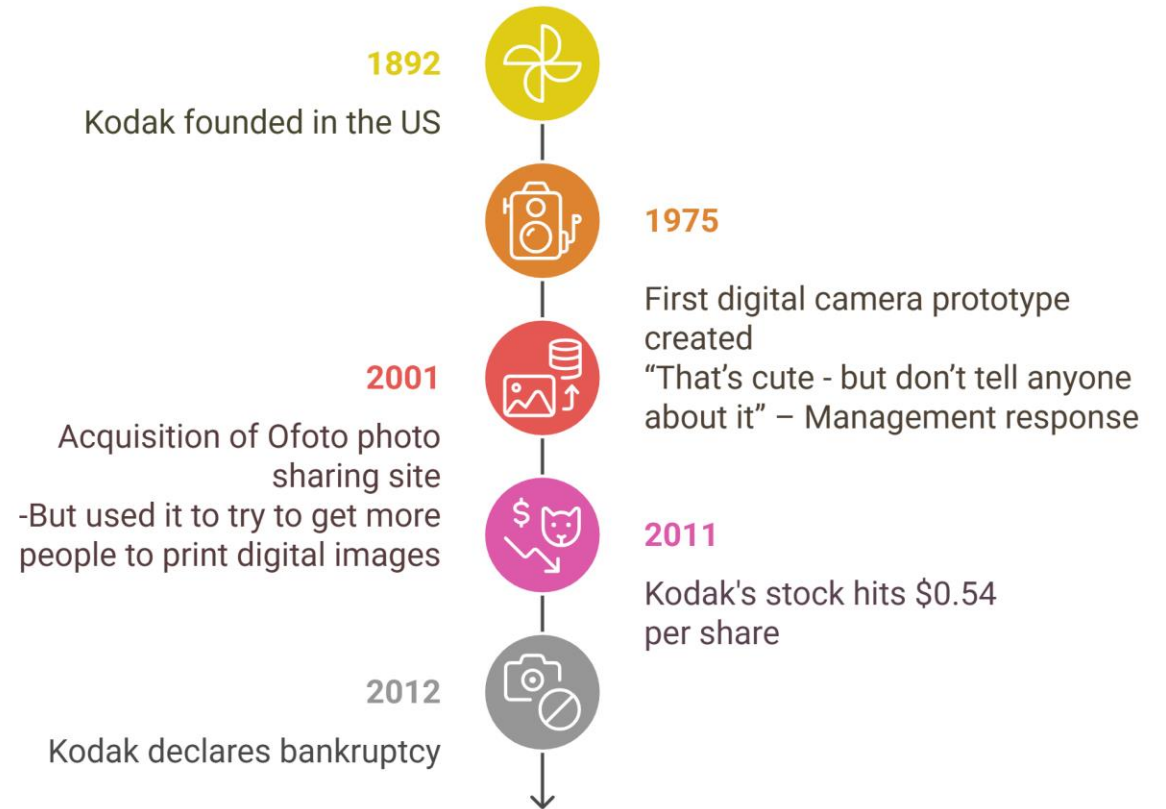
IBM, or International Business Machines Corporation, is a multinational technology and consulting company known for its pivotal role in the development and commercialization of computer hardware and software. Throughout its history, IBM has faced numerous technological disruptions that have reshaped the industry. One of the most significant disruptions occurred in the late 20th and early 21st centuries when IBM transitioned from a hardware-centric business model to a services and consulting-focused organization. This shift was driven by a fundamental change in the company's mental models, allowing IBM to adapt to changing market dynamics and maintain its position as a leading technology company. This case study explores IBM's pivot to consulting and services, examining the

[source](#)

Failing to Adapt

- Kodak's adaptation strategy towards the disruption of digital photography
- Problem
 - Failed to Adapt: focusing too much on their traditional film-based revenue stream.
 - Business Myopia: The company failed to see the disruptive potential of digital technology.

Kodak's Journey Through Digital Disruption



NEXT

L5: IS Posed Challenges &
Issues